

# **Intrusion Detection Systems (IDS): State of the Art and Tradeoffs**

Network Security Report

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# 1 Introduction

Online services have become a critical foundation for modern society, supporting vital services like communication, trading and infrastructure control. This increase in digital dependency has brought about new innovation opportunities but also new cyber threat potential.

Governments and organizations have taken as primary concern the confidentiality, integrity and availability of information systems. In said systems, vulnerabilities can be exploited by malicious agents seeking unauthorized access, financial gain, or system disruption. To mitigate these threats, several defense mechanisms have been developed, such as cryptographic protocols, firewalls, and IDS.

IDS play a crucial role in identifying and responding to security breaches by monitoring network traffic and system behavior. In the past, IDS methodologies had relied on two paradigms:

- **Signature-based detection:** Uses databases of known attack patterns to identify malicious activity.
- **Anomaly-based detection:** Identifies deviations from normal network behavior to uncover previously unseen or novel threats.

However, as cyberattacks become increasingly complex and adaptive, common IDS techniques often struggle to detect zero-day exploits and complex, multi-stage intrusions.

To amend this problem, recent research has focused on enhancing IDS performance through complex analytics, machine learning, and artificial intelligence. These breakthroughs enable systems to automatically learn from network behavior, improve detection accuracy and adapt to newer attack strategies.

This report explores the current state of the art in Intrusion Detection Systems, covering both traditional and modern detection approaches, their tradeoffs and emerging trends in network defense.

## 2 Traditional IDS Approaches

### 2.1 Signature-Based Detection

Signature-based detection (SD), also referred to as Knowledge-based Detection or Misuse Detection [1], operates by identifying malicious activity through pattern matching. In this approach, a signature represents a specific pattern or string corresponding to a known attack or threat. SD systems analyze network traffic and compare captured events against a database of predefined signatures to recognize possible intrusions.[1]

This method leverages accumulated knowledge of previous attacks and system vulnerabilities, allowing for fast and accurate detection of known threats with relatively few false positives. However, its reliance on existing signatures limits its ability to detect novel or modified attacks, emphasizing the need for continuous and timely updates to maintain effectiveness in an evolving threat landscape.

## 2.2 Anomaly-Based Detection

Anomaly-based detection (AD), also referred to as Behavior-based Detection [1], identifies intrusions by detecting deviations from established normal behavior. In this approach, profiles are created to represent expected system or network activities—such as login frequency, processor usage, or email volume—based on continuous monitoring of users, hosts, and network connections over time. These profiles may be static or dynamically updated as system behavior evolves.

During operation, current events are compared against these normal profiles to identify significant deviations that could indicate potential attacks, including attempted break-ins, masquerading, denial-of-service, or Trojan activity [1]. By focusing on behavioral irregularities rather than known signatures, AD can detect new and previously unseen threats, such as zero-day attacks. However, this capability often leads to higher false positive rates, as legitimate but uncommon behavior may also be classified as anomalous.

| Signature-based (knowledge-based)                                                                                                                                                                                                                                                                                           | Anomaly-based (behavior-based)                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Pros</b> <ul style="list-style-type: none"> <li>• Simplest and effective method to detect known attacks.</li> <li>• Detail contextual analysis.</li> </ul>                                                                                                                                                               | <ul style="list-style-type: none"> <li>• Effective to detect new and unforeseen vulnerabilities.</li> <li>• Less dependent on OS.</li> <li>• Facilitate detections of privilege abuse.</li> </ul>                                                |
| <b>Cons</b> <ul style="list-style-type: none"> <li>• Ineffective to detect unknown attacks, evasion attacks, and variants of known attacks.</li> <li>• Little understanding to states and protocols.</li> <li>• Hard to keep signatures/patterns up to date.</li> <li>• Time consuming to maintain the knowledge</li> </ul> | <ul style="list-style-type: none"> <li>• Weak profiles accuracy due to observed events being constantly changed.</li> <li>• Unavailable during rebuilding of behavior profiles.</li> <li>• Difficult to trigger alerts in right time.</li> </ul> |

Figure 1: Pros and cons of intrusion detection methodologies.

[1]

## 3 IDS Types

### 3.1 Network intrusion detection systems (NIDS)

NIDS operate at the network level, where they monitor and analyze traffic to detect attacks or anomalous behavior that may compromise communication infrastructures [2]. Typically deployed on network devices such as routers and switches, they are capable of observing all traffic passing through the network. Their operation generally involves three main stages: monitoring, detection, and response.

In the monitoring stage, transmission-related data such as the number of packets, headers, contents, sizes, port numbers, and established connections are collected. During detection, this data is processed using either signature-based or anomaly-based approaches. Signature-based methods identify known attacks by matching traffic patterns with entries stored in a database, while anomaly-based methods rely on statistical, machine learning (ML), or deep learning (DL) techniques to detect unusual or previously unseen behaviors [2].

The response stage involves notifying system administrators of detected intrusions or, in some cases, dropping or blocking malicious traffic. Since NIDS must process large volumes of data, network sensors are often deployed to handle the scale of traffic, improving the efficiency and effectiveness of intrusion detection [3].

### 3.2 Host intrusion detection systems (HIDS)

Conversely, HIDS are installed directly on individual host devices, where they monitor local system activities and detect malicious behavior within that environment [2]. They primarily collect host-specific data such as system calls, dynamic link library (DLL) files, system logs, and registry keys to identify deviations from normal system operations that might indicate malware, unauthorized access, or resource misuse.

HIDS rely on audit trails gathered from various sources, including operating system logs, application logs, security mechanisms (e.g., firewalls), and network devices such as routers and web servers. They can detect a wide range of malicious activities, including file tampering, segmentation faults, software crashes, unauthorized access attempts, and excessive resource usage [3].

Similar to NIDS, HIDS can employ both signature-based and anomaly-based detection methods [2].

## 4 Modern and State-of-the-Art Approaches

### 4.1 Machine Learning-Based IDS

Machine Learning explores computational models that learn from network data to identify patterns associated with normal or malicious activity. The most common algorithms are decision trees, support vector machines (SVM), k-nearest neighbors (k-NN) and neural networks that classify or predict intrusions based on learned behaviors. These approaches can be labeled as supervised, unsupervised or semi-supervised.

- **Supervised** learning relies on labeled datasets to train models capable of distinguishing between normal and malicious traffic
- **Unsupervised** attempts to identify hidden structure in unlabeled data, often using clustering algorithms, more adaptable for zero-day attack detection
- **Semi-supervised** combines the strengths of both approaches, improving generalization when labeled data is scarce

Each learning approach offers different strengths and weaknesses. Decision Trees and Random Forests provide interpretable results and handle categorical data effectively but may suffer from over-fitting in more complex or noisy datasets. Bayesian Networks are useful in modeling probabilistic relationships and incomplete data while being more computationally intensive. Unsupervised models like k-means clustering are effective at identifying novel attack patterns but they often produce higher false positive rates because they lack labeled training data.

The effectiveness of ML-based IDS is limited by multiple tradeoffs. Although they can adapt to newer threats, their performance depends heavily on the quality of the training data and the appropriate feature selection. Imbalanced datasets, where normal traffic far outweighs malicious samples affects sensitive algorithms leading to biased detection. While ML models can outperform traditional signature based IDS in detecting novel attacks, they may produce higher false positive rates in dynamic network environments [4].

### 4.2 Deep Learning-Based IDS

Deep Learning has emerged as a powerful extension of Machine Learning for intrusion detection systems by automatically learning complex representations from network and data. Vinayakumar *et al.* (2019) proposed a Deep Neural Network (DNN) architecture that outperformed classical ML across multiple benchmark datasets. This study demonstrated that DNN can generalize across different attack types by learning non-linear relationships within network features [5].

Compared to traditional ML models like SVM or Decision Trees, DNN achieved higher detection accuracy while requiring substantial computational resources and extensive hyperparameters tuning.

While DNN can enhance detection capability, challenges emerge in interpretability and scalability for real time intrusion detection. A solution to this was to develop a Scalable Hybrid IDS that demonstrates the potential of deep learning for next-generation IDS solutions [5].

### 4.3 Hybrid and Ensemble Methods

Hybrid and ensemble methods have gained significant traction in recent IDS research due to their ability to combine the strengths of multiple detection paradigms while mitigating individual weaknesses. These systems integrate different algorithms such as ML and RL to enhance detection precision, robustness, and adaptability. By aggregating predictions from different models, hybrid IDS can effectively balance the trade-off between detection accuracy and false positive rates.

Kadam *et al.* (2025) proposed an advanced ensemble model, the Xception Convolutional Generative Adversarial Network (XCGAN), designed specifically for intrusion detection in Internet of Things (IoT) environments [6]. Experimental results demonstrated superior detection accuracy and robustness against sophisticated attacks when compared with standalone deep learning models, validating the benefits of ensemble integration.

Despite their promising performance, hybrid IDS introduce additional computational and architectural complexity. Effective implementation demands careful model synchronization, hyperparameter optimization, and sufficient computational resources challenges that limit their real-time applicability in resource-constrained systems. Nevertheless, ensemble and hybrid approaches remain a key direction in IDS evolution, as they offer a pathway toward more resilient, accurate, and adaptive intrusion detection in dynamic network and IoT ecosystems [6].

### 4.4 AI-Augmented IDS and SIEM Integration

With the development of Artificial Intelligence, traditional IDS have become more adaptive and context-aware systems. IDS are being integrated with Security Information and Event Management (SIEM), enabling real-time event correlation and threat mitigation. Multiple AI techniques like DL and Reinforcement Learning allow systems to learn continuously from network data while maintaining scalability across distributed environments [7].

AI-augmented IDS leverage predictive analytics to correlate alerts, prioritize incidents and automate responses within SIEM environments. This integration closes the gap between detection and response by allowing cross-platform shar-

ing of security intelligence. In particular, Federated learning supports collaborative model updates across multiple edge or cloud nodes without centralized data sharing an approach that enhances privacy and efficiency in large scale infrastructures. Islam *et al.* (2025) demonstrate this paradigm in a distributed UAV security context, where locally trained models communicate aggregated intelligence to improve detection accuracy across the network.

While AI integration greatly enhances adaptability, detection performance and it represents a possible direction for fully automated, intelligent and resilient systems, it also introduces new operational challenges. The reliance on data-driven decision making exposes IDS SIEM systems to risks such as adversarial manipulation and reduced interpretability of AI based outputs. These issues require secure model governance, transparent decision pipelines and continuous validation of learning models.

## 5 Evaluation Metrics

To evaluate the effectiveness of Intrusion Detection Systems (IDS), it is common to use metrics derived from the *confusion matrix*, which summarizes classification results between actual and predicted classes[8].

| Actual / Predicted | Attack              | Normal              |
|--------------------|---------------------|---------------------|
| Attack             | True Positive (TP)  | False Negative (FN) |
| Normal             | False Positive (FP) | True Negative (TN)  |

Table 1: Confusion matrix

Based on these four variables, several metrics can be derived to quantify the detection and misclassification capabilities of an IDS [8][3]:

1. **Classification Rate (CR) or Accuracy:** The ratio of correctly classified instances to the total number of instances.

$$CR = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

2. **Detection Rate (DR) or True Positive Rate (TPR):** The proportion of correctly detected attacks.

$$DR = \frac{TP}{TP + FN} \quad (2)$$

3. **Precision (PR):** The fraction of predicted positive instances that are truly positive.

$$PR = \frac{TP}{TP + FP} \quad (3)$$



4. **Recall:** Measures the proportion of actual attack instances correctly identified by the classifier, it is equivalent to the detection rate (DR).
5. **F-Measure (FM):** The harmonic mean of precision and recall.

$$FM = 2 \times \frac{PR \times DR}{PR + DR} \quad (4)$$

## 6 Comparison and Tradeoffs

Signature-based IDS offer high accuracy and very low false positive rates for known attacks, making them efficient and simple to maintain. However, they lack adaptability, as they cannot detect new or unknown threats without frequent signature updates.

Anomaly-based IDS provide better adaptability by identifying deviations from normal system behavior, enabling the detection of novel attacks. This flexibility, however, results in moderate accuracy, higher false positive rates, and increased computational cost.

Machine Learning-based IDS improve adaptability and accuracy by learning new attack patterns over time. They achieve higher detection performance and lower false positives than anomaly-based systems but require greater computational resources and model tuning.

Deep Learning-based IDS achieve very high accuracy and minimal false positives, showing excellent adaptability to evolving threats. Their main drawback lies in their high computational cost and implementation complexity.

Hybrid IDS combine multiple detection techniques to maximize accuracy and minimize false positives. They demonstrate outstanding adaptability and robustness to new threats but are also the most complex and costly to implement and maintain.

Finally, recent studies indicate that Deep Learning has become the most commonly used approach in IDS research. In the past few years, the majority of proposed systems have been based on DL models, reflecting the growing availability of computational resources such as GPUs and the increasing effectiveness of these techniques in handling complex network data[9].

| Approach         | Accuracy             | False Positives | Adaptability                             | Cost/Complexity |
|------------------|----------------------|-----------------|------------------------------------------|-----------------|
| Signature-Based  | High (known attacks) | Very low        | Poor (cannot detect new/unknown attacks) | Low             |
| Anomaly-Based    | Moderate             | Higher than SD  | Good (detects novel attacks)             | Moderate        |
| Machine Learning | High                 | Lower than AD   | Good (learns new patterns)               | Moderate-High   |
| Deep Learning    | Very High            | Very low        | Excellent (adapts to new threats)        | High            |
| Hybrid/Ensemble  | Highest              | Lowest          | Excellent (combines strengths)           | Highest         |

Table 2: Comparison of different IDS approaches

## 7 Challenges and Future Trends in IDS

### 7.1 Challenges

- **High False Alarm Rates:** IDS often generate excessive false positives, overwhelming security teams and reducing trust in alerts.[3][9][10]
- **Detection of Novel and Zero-Day Attacks:** Many IDS struggle to identify new or unknown attack patterns, especially signature-based systems.[3][9][10]
- **Imbalanced and Outdated Datasets:** Most IDS models are trained on outdated or unrepresentative datasets, limiting real-world effectiveness and generalization.[10]
- **Explainable and Ethical AI:** Incorporating explainable AI to improve trust and regulatory compliance. [11]

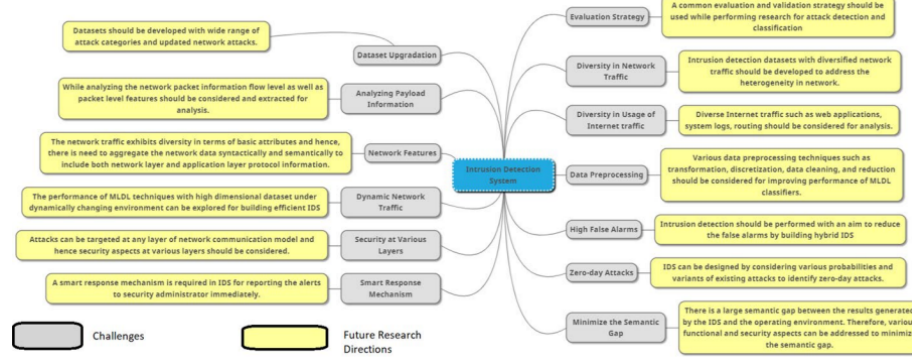


Figure 2: Schematic of challenges and future research directions [3]

### 7.2 Integration with Next-Generation Technologies

Quantum computing has the potential to revolutionize cybersecurity. In the context of Intrusion Detection Systems, it offers several promising approaches to enhance their capabilities, including improved pattern recognition, faster data processing, and more sophisticated security analysis. A prominent application of this paradigm is the Quantum Neural Network (QNN), a next-generation computational model that integrates quantum mechanics into machine learning frameworks to improve threat detection and adaptive security analysis.

In a QNN-based IDS architecture, classical input data are first encoded into quantum states through a Feature Map implemented via quantum circuits.

These encoded quantum representations are then processed within a parameterized quantum circuit—known as the Ansatz—which functions analogously to the weight optimization in classical neural networks. During the training phase, the parameters of the Ansatz are iteratively adjusted to optimize the network’s performance. Finally, the Measurement stage converts quantum information back into classical data, collapsing the quantum states into interpretable outputs.

Through this approach, QNNs can analyze high-dimensional cybersecurity data, detect advanced and evolving threats in real time, and achieve superior performance with reduced computational overhead compared to traditional IDS models.[11]

## 8 Conclusion

IDS have undergone a significant evolution, transitioning from static, rule-based architectures to intelligent, adaptive frameworks driven by machine learning and artificial intelligence. Traditional detection paradigms, such as signature-based and anomaly-based systems, laid the conceptual foundations of intrusion detection but remain constrained by their limited scalability and inability to effectively address emerging threats.

The incorporation of machine learning and deep learning methodologies has substantially enhanced the detection capabilities of IDS, enabling the identification of complex behavioral patterns and previously unseen attacks. However, these advancements have introduced new challenges concerning model interpretability, computational efficiency, and data quality requirements.

Looking ahead, the future of IDS lies in intelligent, collaborative, and continuously learning architectures capable of operating across distributed and heterogeneous environments. This evolution is steering IDS toward autonomous, interpretable, and context-aware security systems, where adaptive intelligence and seamless integration will be essential to counter the growing sophistication and dynamism of modern cyber threats.

Ultimately, as a natural continuation of this study, the next assignment will focus on proposing improvements to existing IDS solutions. In particular, the work will explore the integration of Machine Learning (ML) techniques and the training of intelligent agents using existing datasets[9][10] to enhance detection accuracy and adaptability. The aim will be to develop models capable of identifying and responding to zero-day exploits and other evolving threats, thus contributing to the creation of more resilient and proactive intrusion detection mechanisms.

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## Glossário

**AD** Anomaly-Based Detection. 3, 8

**DL** Deep Learning. 4, 6, 8

**DNN** Deep Neural Network. 5, 6

**HIDS** Host Intrusion Detection Systems. 1, 4

**IDS** Intrusion Detection Systems. 1, 2, 4–10

**IoT** Internet of Things. 6

**ML** Machine Learning. 4–6

**NIDS** Network Intrusion Detection Systems. 1, 4

**RL** Reinforcement Learning. 6

**SD** Signature-Based Detection. 2, 8

**SIEM** Security Information and Event Management. 1, 6, 7

**SVM** Support Vector Machine. 5, 6